MILITARY SPECIFICATION

BELLADONNA ALKALOIDS WITH PHENOBARBITAL ELIXIR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Belladonna Alkaloids with Phenobarbital Elixir (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186	- Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a clear pleasantly flavored elixir containing, in each 5 ml, the following ingredients:

Phenobarbital	16.2000 mg
Hyoscyamine Sulfate	0.1037 mg
Atropine Sulfate	0.0194 mg
Scopolamine (Hyoscine) Hydrobromide	0.0065 mg
Alcohol	23%

3.2 Assay.

3.2.1 Phenobarbital. The elixir shall contain not less than 15.4 mg and not more than 17.0 mg of phenobarbital per 5 ml, when assayed as specified in 4.7.2.

3.2.2 Total alkaloids. The elixir shall contain not less than 0.1166 mg and not more than 0.1426 mg of total alkaloids per 5 ml, when assayed as specified in 4.7.3.

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3.2.3 Alcohol. The elixir shall contain not less than 22.4 percent and not more than 23.6 percent of alcohol per 5 ml, when determined in accordance with the USP method for "Alcohol Determination." As an alternate, the Gas-Liquid Chromatographic Method shall be employed.

3.3 pH. The pH of the finished elixir shall be not less than 4.0 and not more than 5.5 when determined potentiometrically at 25° C using the USP method.

3.4 Palatability. The elixir shall be palatable and pleasant to the taste with no unpleasant aftertaste. Not later than the time specified for opening of bids or receipt of proposals, the offeror shall submit to the contracting officer six (6) individually packaged, 4 fl oz bottles of the elixir, representative of the product which the offeror proposes to furnish. Two (2) samples will be subjected to panel testing for a determination of palatability (see 4.7.1). The remaining samples will be used by cognizant Government inspection and quality assurance activities for determining compliance of supplies furnished hereunder with the palatability requirement. Approval of palatability of any sample submitted by the offeror will not constitute approval of the sample as to any other requirement of this specification. The requirement for submission of samples for use in determining compliance with the palatability requirement may be waived, provided the offeror states, in his bid or proposal, that the product he proposes to furnish is the same product he has offered to the purchasing activity on a previous procurement and the contracting officer determines that such product was previously procured and/or tested by the purchasing activity and found to comply with the palatability requirement.

3.5 Ingredients.

3.5.1 Alcohol, atropine sulfate, phenobarbital, and scopolamine (hyoscine) hydrobromide. The alcohol, atropine sulfate, phenobarbital and scopolamine (hyoscine) hydrobromide used in the manufacture of the elixir shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.5.2 Hyoscyamine sulfate. The hyoscyamine sulfate used in the manufacture of the elixir shall comply with the tests, standards, and requirements of the N.F., including any supplements or revisions thereto.

3.5.3 Other ingredients. All other ingredients used in the preparation of the elixir shall be of USP or NF quality, or if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade.

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3.6 Delivery. Not more than 6 months shall have elapsed from the date of manufacture of the elixir to the date of delivery to the Government.

3.7 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated item, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of elixir produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.2.1 Inspection. The contractor shall notify the Food and Drug Administration (FDA) of the anticipated dates for (a) testing of each of the ingredients entering into the finished product (b) the manufacturing of the product, and (c) the final testing, packaging, and packing. Such notification to the FDA shall be submitted so that it is received in the FDA office not less than 10 calendar days before each facet of testing and manufacture.

Each lot of each ingredient entering into the final product shall be tested by, or witnessed by, the FDA quality assurance representative, and it shall be determined by the quality assurance representative that each of the ingredients complies with designated requirements.

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Sales of documents ... the following documents ... of the issue ...

The quality assurance representative shall witness the weighing or measuring of each ingredient in the manufacture of the finished product and shall attest that the weight or measure of each ingredient is 100 percent of that called for in the formula for elixir.

The quality assurance representative shall witness the compounding, manufacturing, and bottling of the finished product.

Notwithstanding the presence of the quality assurance representative during the above testing, manufacturing, and inspection procedures, the contractor is not relieved of his obligation to provide material in full conformance with requirements of the contract; and the contractor will remain responsible for any failure to so meet such requirements.

4.2.1.1 As an alternate to the above Inspection, the bidder shall submit a method of assay, for determining the quantity of each of the active ingredients in the tablets, for approval and acceptance by the Government. If such method is approved and accepted by the Government, the material offered shall be tested by such method and shall comply with the requirements established by the Government for the method submitted.

4.2.1.2 Those ingredients which have been tested and accepted by the contractor, prior to award, need not be tested by, or witnessed by, the quality assurance representative. In lieu thereof, the quality assurance representative may accept protocols and perform spot testing to verify the quality of the ingredients, or, protocols, standards, and samples of the ingredients may be submitted by the quality assurance representative to the Defense Personnel Support Center for appropriate testing.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

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TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay			
Phenobarbital	3.2.1	S-2	5 } Composite
Total alkaloids	3.2.2	S-2	5 } Composite
Alcohol	3.2.3	S-2	5 } Composite
pH	3.3	S-2	5 } Composite
Palatability	3.4	---	---

1/ Where possible, the sample shall be used for two or more tests.

4.4 Examination. The finished elixir shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Elixir not free from foreign matter.
102	Elixir has crystal formation.
103	Elixir not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the unit of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

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4.6.1 Records of examinations and tests performed by or

4.7 Tests.

4.7.1 Palatability test. A taste panel consisting of 10 members will be used to determine acceptability of samples. Samples will be prepared for testing, coded, and served to panel members under controlled serving conditions, e.g., all samples will be of the same amount, and served at the same temperature; each panel member will receive an equal number of samples; the order of serving will be varied among panel members; an interval of at least five (5) minutes will elapse between successive samples and panel members will rinse their mouths with water (room temperature) after each sample; panel members will test without interference either from each other or from outsiders. The product offered shall be rated equal to or better than the palatability standard 1/, when determined by the taste panel, using the following 9-point hedonic rating scale. The average rating of the sample shall be equal to or greater than the average rating of the standard, similarly prepared and tested.

1	2	3	4	5	6	7	8	9
Dislike extremely	Dislike very much	Dislike moderately	Dislike slightly	Neither like nor dislike	Like slightly	Like moderately	Like very much	Like extremely

1/ The palatability standard is available, upon separate request to the contracting officer, from the Defense Personnel Support Center, Directorate of Medical Materiel.

4.7.2 Phenobarbital assay. The phenobarbital content in the elixir shall be determined as follows:

Reagents and Standard.

Reagents.

pH 9.5 buffer solution.

(A) 0.1N Boric acid. Dissolve 6.18 grams of Boric Acid in a liter of purified water.

(B) 0.1N Sodium hydroxide. Dissolve 4.00 grams of Sodium Hydroxide pellets in a liter of purified water. Mix 680 ml of Solution (B) with 1000 ml of Solution (A) and dilute to 2000 ml with purified water.

Standard preparation.

Weigh accurately approximately 81 mg of USP Phenobarbital reference standard into a 100 ml volumetric flask. Dilute to volume with SDA-3A alcohol, and mix well. Pipet 1 ml into a 100 ml volumetric flask and dilute to volume with buffer soltuion.

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Sample preparation.

Pipet 5 ml of elixir into a separatory funnel. Extract with four 25-ml portions of chloroform. Combine the extracts in a second separatory funnel and filter through chloroform wetted filter paper into a 100-ml volumetric flask. Dilute to volume with chloroform. Pipet 5 ml of the chloroform extract into a beaker and evaporate to dryness on a steam bath. Add one ml of SDA-3A alcohol to the residue and using pH 9.5 Buffer Solution, quantitatively transfer the residue to a 100 ml volumetric flask. Dilute to volume with the Buffer Solution and mix well.

Using 1 cm silica cells, determine the absorbancies of standard and sample, respectively, in a suitable spectrophotometer versus a blank of 9.5 Buffer Solution, at the point of maximum absorbance of about 300 nm.

Calculation:

$$\frac{\text{Absorbance of Sample}}{\text{Absorbance of Standard}} \times 16.2 = \text{mg phenobarbital per 5 ml.}$$

4.7.3 Total alkaloid assay. The total alkaloid content of the elixir shall be determined as follows:

Standard.

From alkaloids previously dried at 100° C., accurately weigh 1.037 gram of NF Hyoscyamine Sulfate reference standard, 194.0 mg of USP Atropine Sulfate reference standard, and 65.0 mg of USP Scopolamine (Hyoscine) Hydrobromide reference standard. Quantitatively transfer to a 1000 ml volumetric flask, dissolve in and dilute to volume with purified water. Transfer exactly 10 ml to a 100 ml volumetric flask and make to volume with purified water. This solution contains 0.1296 mg of total alkaloids per ml.

Reagents.

Phosphate buffer. Dissolve 38 grams of monobasic sodium phosphate and 2 grams of anhydrous dibasic sodium phosphate in, and dilute with purified water to make 1000 ml. Adjust the pH, if necessary, to 5.3 ± 0.1.

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Bromocresol purple solution. Dissolve 400 mg of bromocresol purple in 30 ml of ethyl alcohol. Add 6.3 ml of 0.1N sodium hydroxide and dilute with purified water to 500 ml.

Buffer-dye mixture. Mix equal portions of phosphate buffer and bromocresol purple solution in a separatory funnel and extract with chloroform until the chloroform extract is colorless. Discard the chloroform.

Sample preparation.

Pipet 25 ml of elixir into a separatory funnel. Add 10 ml of Buffer-Dye Mixture. Shake. Extract with two 50 ml portions of Chloroform, combining the extracts by filtering into a 100 ml volumetric flask. Prepare a standard by transferring 5 ml of the standard preparation into a separatory funnel and proceed as described in "Sample Preparation," above, beginning with "Add 10 ml of Buffer-Dye Mixture . . . ." Using 1 cm silica cells, obtain the absorbancies of the standard and sample preparations at 410 nm in a spectrophotometer versus a reagent blank.

Calculation:

$$\frac{\text{Absorbance of Sample}}{\text{Absorbance of Standard}} \times (0.1296) = \text{mg total alkaloids per 5 ml.}$$

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3), and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle containing 1 gallon (3.78 liters), as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available immediate container considered acceptable by the Food and Durg Administration (FDA) for the product container therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit package. Each unit, one bottle, shall be packaged as specified in 5.2.10 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Four unit packages shall be packed in an exterior container as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Four unit packages shall be packed in an exterior container as specified in 5.4.3.4 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the date of manufacture prefixed by "MFD".

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.

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5.4.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129, except that the date of manufacture shall be shown in lieu of the date packed. Lot (control) number and the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30° C (59° - 86° F)" shall also be shown.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item(s):

National Stock Number	Item Identification
6505-00-871-8307	BELLADONNA ALKALOIDS WITH PHENOBARBITAL ELIXIR, 1 GAL (3.78 liters)

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1632

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DOCUMENT IDENTIFIER AND TITLE

MIL-B-37474 - BELLADONNA ALKALOIDS WITH PHENOBARBITAL ELIXIR

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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